

Daewon Kim

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<https://github.com/vinyl810>

EDUCATION

KAIST, Daejeon, Republic of Korea, *M.S.* (2025.09 – Current)

- Major in Culture Technology
- Affiliated with *Visual Cognition Lab* (advised by *Professor: Jeongmi Lee*)
- Cumulative GPA: 4.20/4.30

KAIST, Daejeon, Republic of Korea, *B.S.* (2018.02 – 2025.08)

- Major in Computer Science
- Double Major in Bio and Brain Engineering
- Semiminor in Artificial Intelligence
- Cumulative GPA: 3.19/4.30

Chungnam Science Highschool, Gongju, Republic of Korea, (2016.03 – 2018.02)

- Early graduation in 2018

PUBLICATIONS

Song, T., Min, S., **Kim, D.**, Won, S., Go, G., Cho, S. (2026). *The Influence of Album Arts on the Perception of Musical Emotion*. Poster presented at the Annual Conference of the Korean Society for Cognitive and Biological Psychology, Republic of Korea.

Kim, D., Min, S., Lee, K., Choi, Y. (2026). *Adaptive Tarot Cards: A Generative AI-based Personalized Symbolic System Evolving Over Time*. Creative Award, HCI Korea 2026, Hongcheon, Republic of Korea. (*equal contribution*)

WORK EXPERIENCES

KOAST Inc., Seoul, Republic of Korea, *Web Graphic & Frontend Engineer* (2023.01 – 2025.02)

- Tech Stack
 - WebGL & Shaders
 - CesiumJs
 - OpenLayers
 - Flutter
 - React, Vue
- Served as an industrial technological agent for Korean Military
- Developed oceanic digital twin web application for government agencies
- Developed 2D & 3D GIS web applications for Korea Meteorological Administration, Korea Coast Guard etc.
- Developed oceanic and atmospheric SPA web applications

PROJECT EXPERIENCES

Safe Sustained Disagreement: Cognitive Reframing by Spectating a Non-Convergent Debate Among Extreme Emotion Personas(KAIST GCT79900) (2026.03 – 07)

- Conducted study on a system designed to cognitively reframe everyday concerns by spectating a debate done by agents with emotionally extreme personas

- Performed SFT & WDPO on Gemma-4, synthesized chat samples, developed safety gates for the personas to remain safe and to not drift to hopelessness
- Conducted user study on the influence of the system with 23 participants, and analyzed the data
- Contributed to the design of the project, the design and architecture of the system, the design of the user study, and the final analysis of the user study and performance

Analysis of Neural Dynamics of Alpha Power During Simulated Driving Tasks(KAIST BiS427) (2025.05 – 07)

- Conducted EEG-based analysis of sustained attention and fatigue during simulated driving tasks, comparing prolonged and shorter driving datasets
- Demonstrated that prolonged driving induces declining alpha power, higher entropy, reduced synchrony, and unstable neural states, suggesting EEG alpha metrics as robust indicators for neuro-ergonomic driver monitoring systems
- Engineered signal processing pipeline: alpha-band (8-13 Hz) filtering, dimensionality reduction (PCA, t-SNE), and temporal sliding-window correlation
- Contributed to analysis utilizing python libraries SciPy, scikit-learn

Oceanic Digital Twin Project(KOAST Inc.), *Web Graphic & Frontend Engineer* (2024.04 – 2025.02)

- Developed 3D GIS web application which models oceanic environment as a ‘digital twin’ of the real world
- Contributed to engineering 3D WebGL GIS layers modeling oceanic and atmospheric environments such as oceanic wave heights, water temperature, and chlorophyll density etc.
- Contributed to frontend architecture by developing the main 3D framework, menus etc.
- Tech Stack: CesiumJs, WebGL, React

Tideland Navigation Application Project(KOAST Inc.), *Web Graphic & Frontend Engineer* (2024.04 – 2025.02)

- Developed 2D GIS mobile application which helps users in tideland to find safe spots between the interchange of short and high tides
- Contributed to engineering 2D GIS layers modeling terrains and the danger/safe spots during high tides
- Contributed to mobile application architecture by developing the boilerplate of the flutter app
- Contributed to frontend architecture by developing the Webview part of the app in React
- Tech Stack: OpenLayers, React, Flutter

KMA Oceanic Web System Project(KOAST Inc.), *Web Graphic & Frontend Engineer* (2023.03 – 2024.10)

- Developed two GIS web applications for oceanic and atmospheric information (publicly available application: [link](#))
- One application is for public use, and the other one is for oceanic researchers and experts
- Contributed to frontend architecture by developing the main page, submenus, the GIS layers, and varieties of features etc.
- Contributed to engineering 2D GIS layers modeling oceanic and atmospheric phenomena such as typhoon, heat index, gust, and ripple current etc.
- Tech Stack: OpenLayers, WebGL, Vue

Korea Coast Guard AI Search and Rescue Project(KOAST Inc.), *Web Graphic Engineer* (2023.12 – 2024.08)

- Developed GIS web application for assisting search and rescue happening in seas based on AI
- Contributed to engineering 3D WebGL GIS layers modeling oceanic and atmospheric phenomena and environment such as wind, oceanic current, and salinity etc.
- Tech Stack: WebGL, OpenLayers

KMA Atmospheric Observation Device Control System for Researchers Project(KOAST Inc.), *Frontend Engineer* (2024.01 – 2024.08)

- Developed a web application for the researchers in the field of atmospheric science, providing the features of handling and monitoring the atmospheric observation devices
- Contributed to frontend architecture by developing the boilerplate of the project and reviewing the code
- Tech Stack: OpenLayers, React

Smart Fish Farm Project(KOAST Inc.), *Frontend Engineer* (2023.04 – 2024.03)

- Developed fish farm managing web application for fish farm administrators
- Contributed to frontend architecture by developing the main page, visualizing the fish tanks, drawing svg graphs, and displaying the CCTV screens etc.

- Tech Stack: React, Vue

Duplicating PatchCore Project(KAIST CS470) (2022.03 – 06)

- Conducted a small project duplicating the [PatchCore](#) paper and test the effect of background anomaly removal preprocessing in computer vision to improve the output metrics of the original paper
- Contributed to code reproduction of the original paper based on PyTorch and utilizing libraries such as sklearn and scipy
- Tech Stack: PyTorch

Music Education Web Application Project(KAIST CS374), *Frontend Engineer* (2021.03 – 06)

- Developed music education web application for novice music sheet readers ([github link](#))
- Contributed to frontend architecture by developing interactive music sheet, player etc.
- Tech stack: React

Analyzing Interests for Celebrities Using Internet News Project(KAIST CS492) (2020.03– 06)

- Conducted a small project analyzing internet news and the comment section to analyze public interest and emotion for Korean celebrities, consequently creating a word cloud consisting of the keywords that represent each celebrity to the public audience
- Contributed to analyzing the result by applying k-means clustering, and in calculating the negative ratio to subject celebrity etc.
- Tech Stack: PyTorch, R

RELATED COURSEWORKS

Culture Technology

Introduction to Culture Technology | Cognition and Emotion | Human Visual Cognition | Digital Architecture | Natural Language Processing for Culture Technology | Computer Graphics for Culture Technology

Computer Science

Calculus | Introduction to Algorithms | Natural Language Processing with Python | Introduction to R for Data Science | Discrete Mathematics | Programming Principles | Introduction to Human-Computer Interactions | Machine Learning | Introduction to Artificial Intelligence | Introduction to Database | Machine Learning for 3D Data | Programming for AI

Bioengineering

Bioengineering Fundamentals | Bio-Information Processing | Bio-Data Structures | Molecular and Cellular Biology | Anatomy and Physiology | Computational Neuroscience | Brain Science Fundamentals

REWARDS AND HONORS

New Challenge Award, Creative Award Track, HCI Korea 2026 — for *Adaptive Tarot Cards: A Generative AI-based Personalized Symbolic System Evolving Over Time*

INVOLVEMENT EXPERIENCES

KAIST Graduate School of Culture Technology Student Council, *Vice President* (2026.02 – current)

- Served as a Vice President of the Student council, responsible of executing various programs regarding the students of GSCT.

KAIST Graduate School of Culture Technology DaVinci Lab, *Teaching Assistant* (2026.03 – 06)

- Worked as a teaching assistant for the DaVinci Lab program, providing aids and advices for the first year graduate students of GSCT.

Korea National Science Museum STEAM Camp, *Mentor/Head Mentor* (2021, 2022)

- Participated as a mentor in the STEAM Camp program targeting elementary and middle school student.
- Participated as a head mentor in year 2022, organizing courses, mentors, and administrative works.